GAPPS user guide

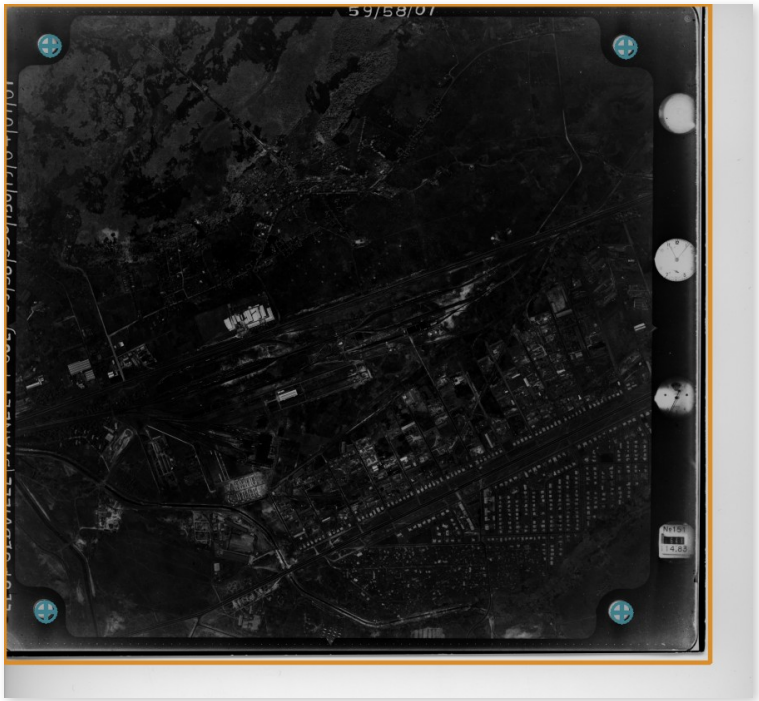
From scanned aerial photographs to images ready for photogrammetry: the steps, why they are needed, and which settings to use



5958_001 after GAPPS: cropped, reprojected on its fiducial marks, 900 dpi, sharpened

GeoRiskA Airphoto Pre-Processing Suite · RMCA / VUB · Dev_202610

Why scans need pre-processing



Raw scan 5958_001 (13600 × 12598 px, 1600 dpi). Orange: the photo frame kept by Tool 1. Blue: the 4 fiducial marks (FMs).

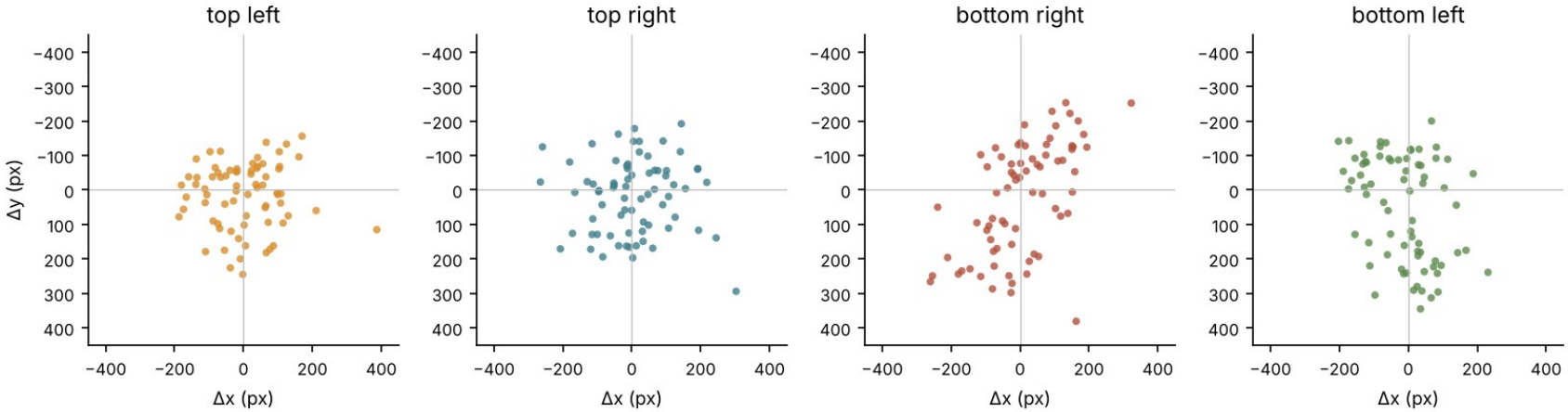
1. **Every scan is different**
   Film placed by hand: each scan has its own size, offset and small rotation, plus the white scanner bed.

2. **The camera geometry is in the FMs**
   The FMs are exposed by the camera itself. Their centre is the reference for the principal point.

3. **Photogrammetry wants identical frames**
   Same size, FMs at the same pixels, principal point at the image centre. Then one camera calibration fits all images.

4. **1600 dpi is heavy**
   About 300 MB per scan. A lower, sharpened resolution is faster to process.

Marks move by hundreds of pixels

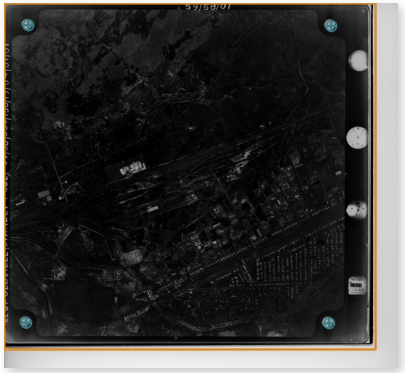


Position of each FM in the 72 cropped scans of box 37582B01, relative to its median position. The top-left mark alone spans 574 px in x and 401 px in y.

This is why GAPPS reprojects each image on its own FMs. Cropping alone would leave the principal point at a different pixel in every image, and the photogrammetric software would see a different camera each time.

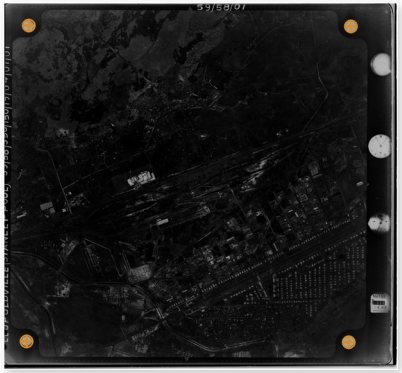
The workflow in four images

1 • Crop Tool 1



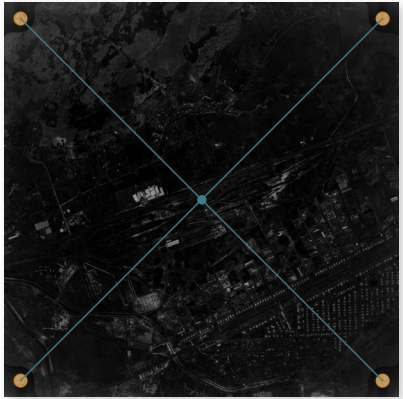
`A_CanvasSized_Cropped`

2 • Find the FMs Tool 2



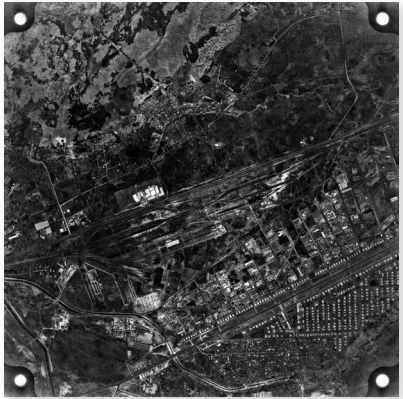
`Out_fiducialmarks.csv`

3 • Reproject Tool 3



`B_Reprojected`

4 • Resample Tool 4



`C_Resampled`

Same photo at each step. Step 3 moves the 4 FMs (orange) to the positions set by the camera model; the diagonals then cross at the image centre.

Tool 5 then makes one mask that hides the FMs in the corners during the photogrammetric processing. Tool 2 also includes a review step (2e) to correct doubtful marks.

Set up and launch

1. **Install once**
   Clone GeoRiskA/GAPPS, create .venv inside the GAPPS folder, then pip install -r requirements.txt.

2. **Point GAPPS to your scans**
   Edit GAPPS_config.json: the folder with the raw scans and the folder for the results.

3. **Launch**
   Launch_GAPPS_Win.bat (Windows), Launch_GAPPS_Mac.command, or run_gapps.py from the .venv.

4. **One output folder per batch**
   Use one results folder per camera / scanning batch: every tool reads and writes there.

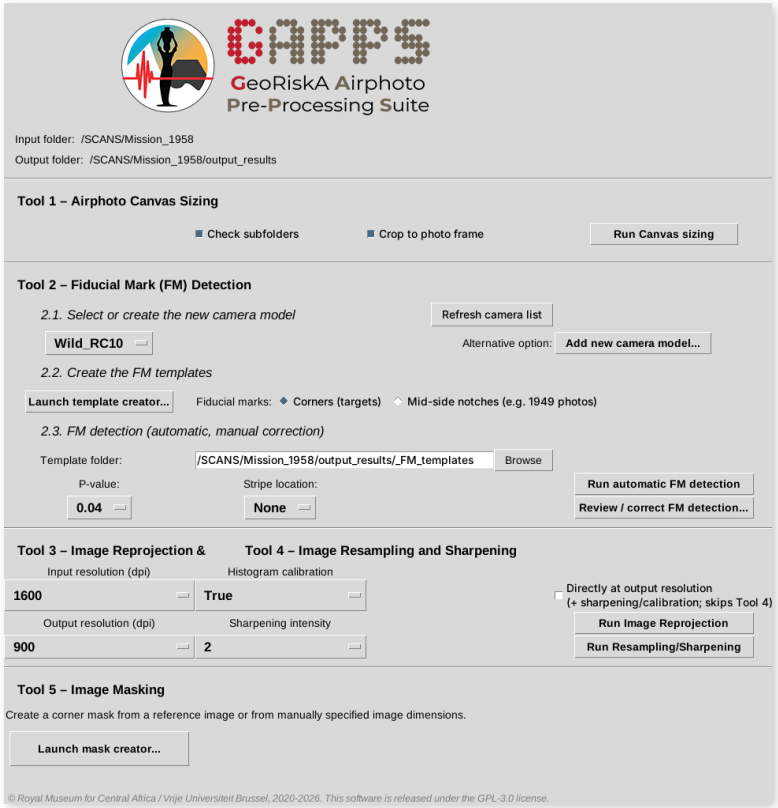
GAPPS_config.json

```
{
 "raw_scanned_airphoto_folder_path":
    "G:\\KIN_SCANS\\37582B01",
 "output_results_folder_path":
    "G:\\KIN_SCANS\\37582B01\\output_results"
}
```

Results folder after a full run

```
output_results/
  A_CanvasSized_Cropped/   cropped scans
    Out_fiducialmarks.csv   FM pixels
    corners/_all_fiducials/ check figures
  _FM_templates/           templates
  B_Reprojected/           1600 dpi
  C_Resampled/             900 dpi, final
```

The main window follows the workflow



1. **Tool 1**
   Crop the scans to the photo frame.

2. **Tool 2.1**
   Pick the camera model (or add one).

3. **Tool 2.2**
   Choose Corners or Mid-side notches, then make the templates.

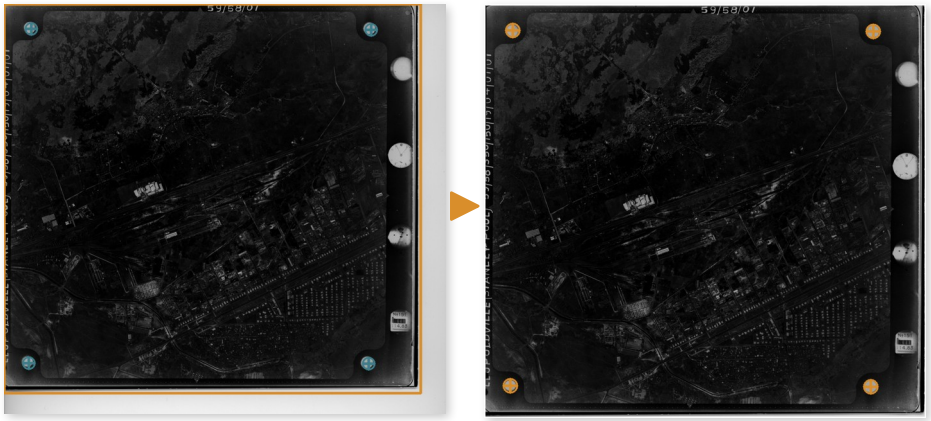
4. **Tool 2.3**
   Run the detection, then Review / correct.

5. **Tools 3 + 4**
   Set resolutions and sharpening, run the reprojection.

6. **Tool 5**
   Make the corner mask.

Run the tools top to bottom. The settings shown are the recommended defaults (1600 → 900 dpi, calibration on, sharpening 2).

Tool 1 · Crop to the photo frame



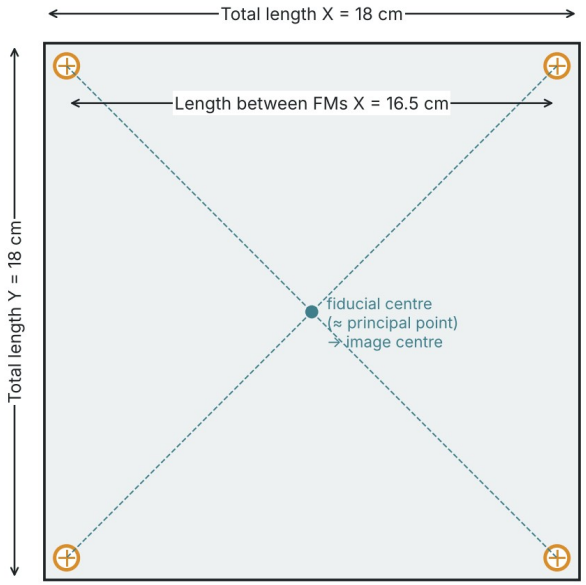
Left: raw scan with the white scanner bed. Right: cropped to the black film frame (12823 × 11965 px).

Parameter	Use	Why
Check subfolders	On	Finds scans in sub-folders (e.g. Bande_01, Bande_02). Names must be unique.
Crop to photo frame	On	Removes the scanner bed, so the FM search starts on the film.

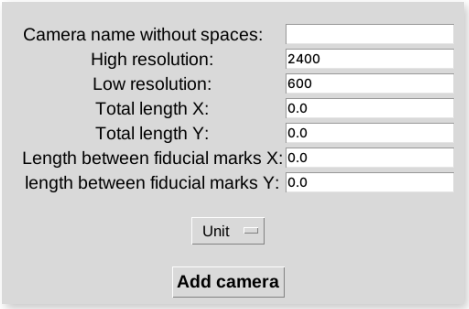
Safe to re-run: images already cropped are skipped, so an interrupted run just continues. Scans where no frame is found are listed in __error_images.txt.

Off "Crop to photo frame": scans are only padded to a common canvas (A_CanvasSized). Use it only for scans already cut to the film.

Tool 2.1 · Camera model: FM targets



Wild RC5a: 18 × 18 cm frame, FMs 16.5 cm apart. GAPPS turns this into pixels for each dpi.



"Add new camera model..." window

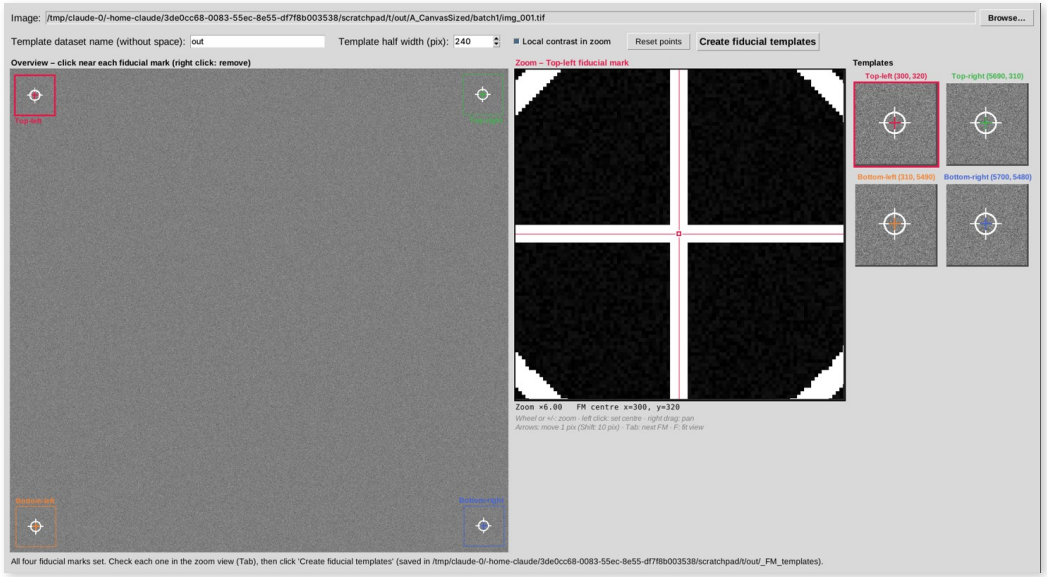
Enter from the camera calibration report:

- Total length X / Y of the frame
- Distance between the FMs in X / Y
- Unit (cm or inch) and dpi range

Parameter	Use	Why
1600 dpi	**11339 px**	Image size; FMs at 472 and 10866 px
900 dpi	**6378 px**	Image size; FMs at 266 and 6112 px

Pixel targets of the Wild RC5a model (from Wild_RC5a_Airphoto_dimensions_vs_dpi.csv). Not needed for mid-side notches: their targets are learnt from the images.

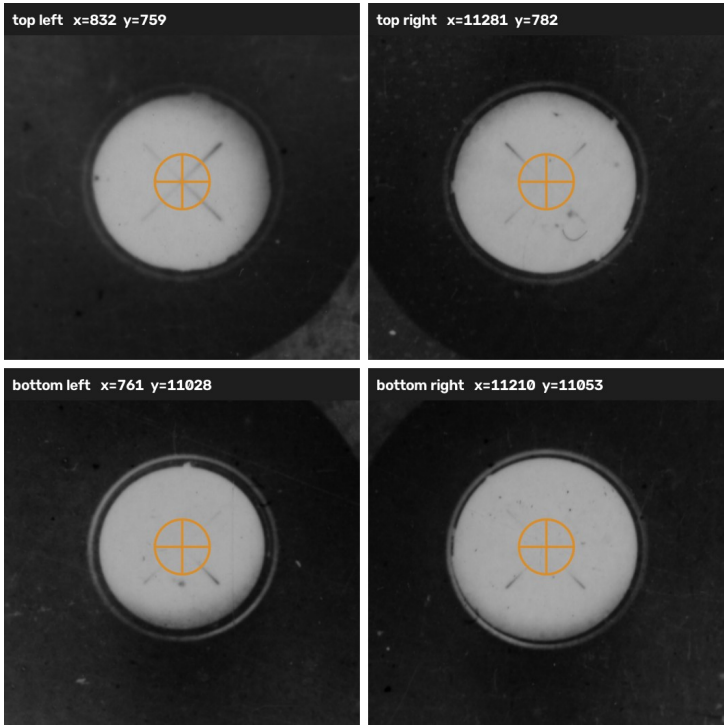
Tool 2.2 · Templates of the marks



Template creator (test image). Click near each mark in the overview, then click its exact centre in the zoom.

Parameter	Use	Why
Image	**A sharp, typical scan**	Every image is matched against these 4 templates.
Centre	**Pixel-exact**	An offset here shifts every detected FM.
Half width	**240 px (corners), 250 px (notches)**	Mark plus some frame. Less if marks are close to other features.
Sets	**One per camera / batch**	Several sets can live in _FM_templates: the best match wins.

Tool 2.3 · Automatic FM detection

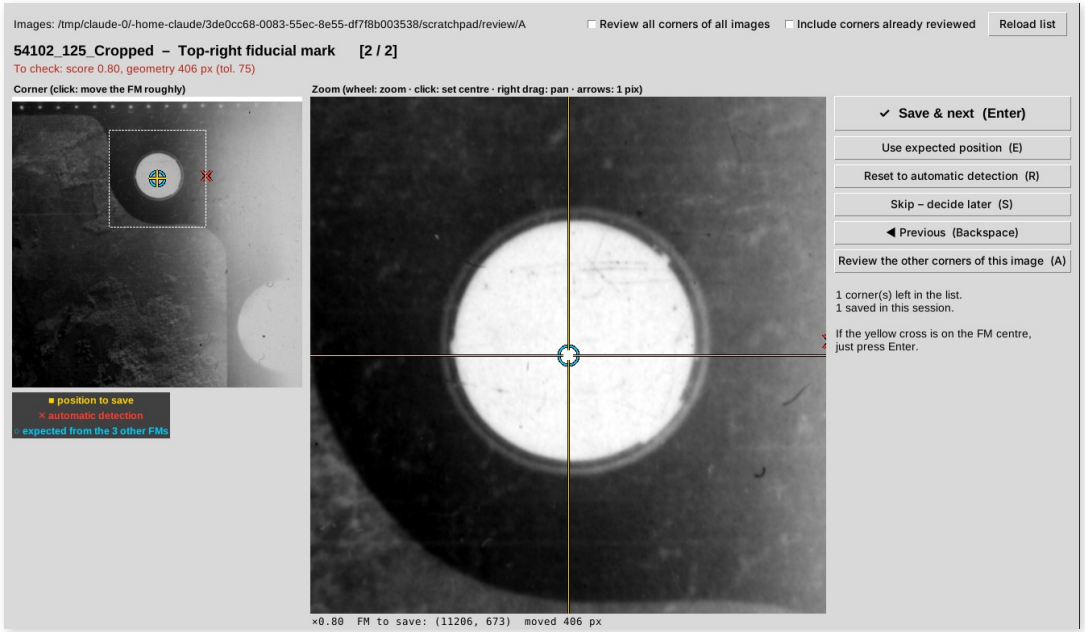


Detected FM centres on 5958_001 (720 px windows). After a run, check the overview figures in corners/_all_fiducials.

Parameter	Use	Why
Template folder	_FM_templates	Filled in by default after creating the templates.
Stripe location	None	Only if a black no-data stripe is left on one side (uncropped scans).
P-value	0.04	With a stripe: moves the search window inward by P × image size on that side.
Fiducial marks	Corners	Mid-side notches for photos like the 1949 Leopoldville set.

How it works: each corner of the image (2500 px window) is matched with the templates. A score below 0.88 or a 4-mark shape that differs from the rest of the batch sends the image to the review list.

Tool 2e · Review and correct: always run it



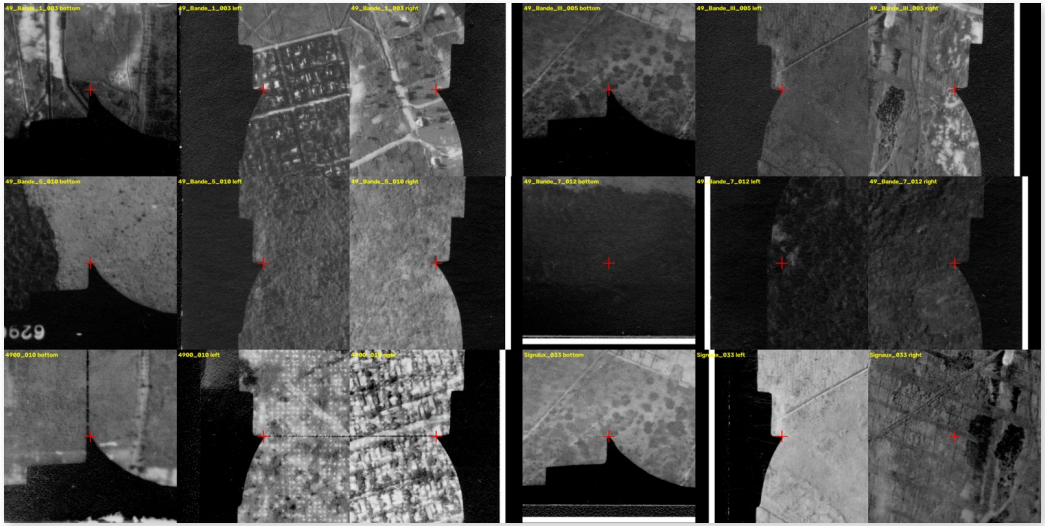
A mark detected 406 px off (red ×). Click the true centre in the zoom; the blue circle is where the other 3 marks expect it.

Only the doubtful marks are shown: low score, wrong geometry, or missing.

Enter correct: save and next
click set the centre (wheel: zoom)
E use the expected position
N mark not visible (notches)
S / R skip / reset

Saved at once in Out_fiducialmarks.csv (backed up first). Then go to Tool 3.

Special case · Mid-side notches (1949 photos)



Detected notch tips (red +) at the bottom, left and right of six 1949 images.

Why a different method: a notch is just a cut in the frame edge, so GAPPS matches the shape of the dark frame, not the grey values next to it.

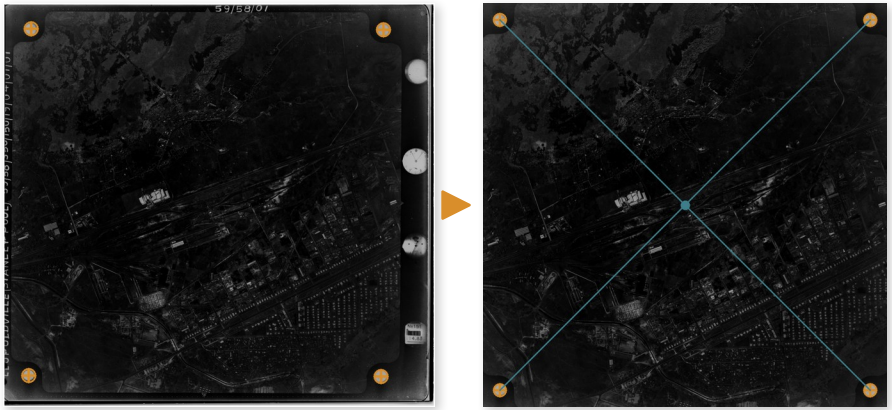
1. **Select "Mid-side notches"**
   In Tool 2.2. The choice is used by every later step.

2. **Templates on the notch tips**
   Half width 250 px. Skip a side that is outside the scans (3-notch set).

3. **Detect, then review**
   GAPPS learns the notch layout and predicts missing ones. Press N if a notch is not visible.

4. **Reproject**
   No camera model needed. An affine fit on ≥ 3 reliable notches.

Tool 3 · Reprojection on the fiducial marks

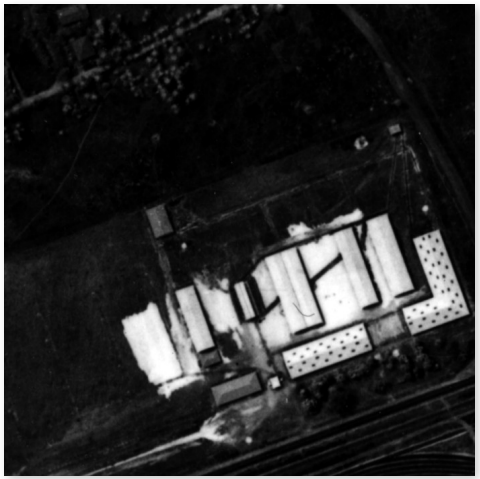


The detected FMs (left) are mapped onto the camera-model targets (right): 11339 × 11339 px, FM diagonals crossing at the centre.

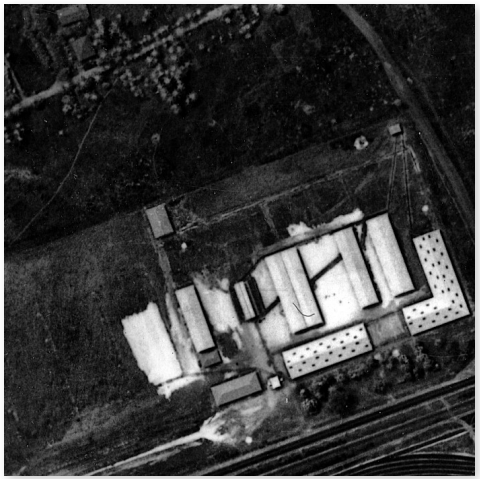
Parameter	Use	Why
Camera model	**Your camera**	Gives the target FM pixels.
Input resolution	**Scan dpi (1600)**	Must match the scans, or the targets are wrong.
Directly at output resolution	**On**	Tools 3 + 4 in one step: ~2× faster, no 230 MB intermediate per image.

Leave "Directly" off only if you need the full-resolution B_Reprojected images. Then run Tool 4 afterwards.

Tool 4 · Resample and sharpen



Plain resize to 900 dpi



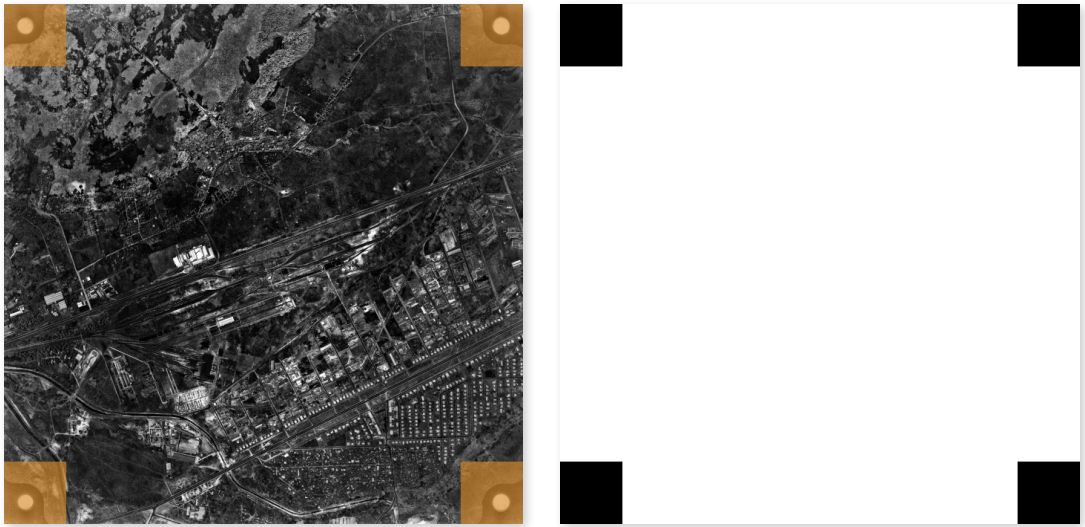
GAPPS: sharpening 2 + calibration

Same 700 × 700 px area of 5958_001 at 900 dpi. Calibration lifts detail in dark and bright areas, sharpening restores edges.

Parameter	Use	Why
Output resolution	900 dpi	Detail vs. processing time; 28 µm pixels.
Histogram calibration	True	Local contrast (CLAHE): more tie points in shadows.
Sharpening intensity	2	0 none · 1 low · 2 medium unsharp mask.

Keep the same settings for every batch of one project, so all images look alike to the photogrammetric software.

Tool 5 · One mask to hide the marks



Left: 12 % corners on a final image. Right: the mask (black = ignored).

1. **Why**
   FMs are at the same pixels in every image. Tie points on them would fit a fake "static" scene.

2. **Reference image**
   Pick one image from C_Resampled: the mask takes its size.

3. **Corner size**
   12 % by default. Increase until the marks and their plates are fully covered.

4. **Use**
   Load the same mask for all images in your photogrammetry software.

Recommended settings at a glance

Parameter	Use	Why
Tool 1	**Subfolders on · Crop to frame on**	Removes the scanner bed; resumes safely.
Tool 2.1	**Camera of the mission**	Defines the FM target pixels (corners only).
Tool 2.2	**Corners or Mid-side notches · 240 / 250 px**	One template set per camera or batch.
Tool 2.3	**Stripe None · P 0.04**	Stripe only for uncropped scans.
Tool 2e	**Always review the list**	One wrong mark distorts the whole image.
Tools 3 + 4	**1600 → 900 dpi · Directly on · CLAHE True · Sharpening 2**	Fast, identical geometry, crisp images.
Tool 5	**Corners 12 %**	Hides the marks from the tie-point search.

If something fails, look here first: __error_images.txt (Tools 1 and 3), __FM_detection_errors.txt and the corners/_all_fiducials figures (Tool 2).